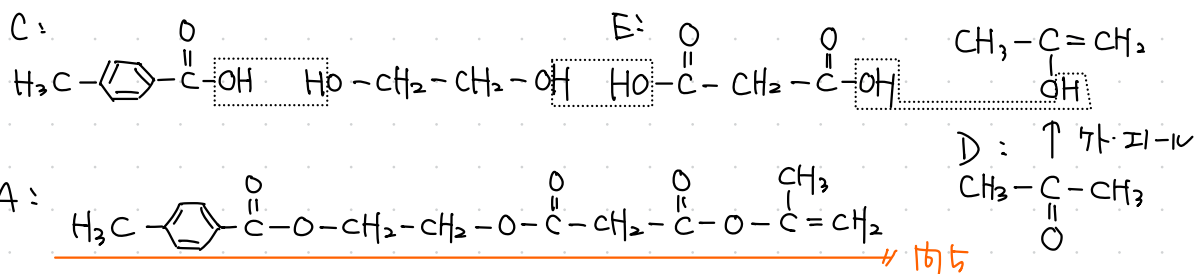
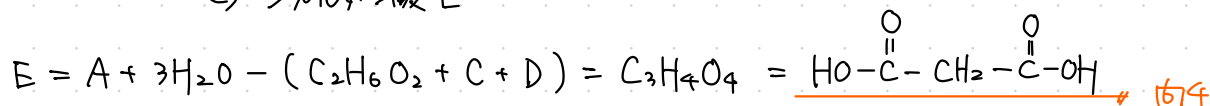
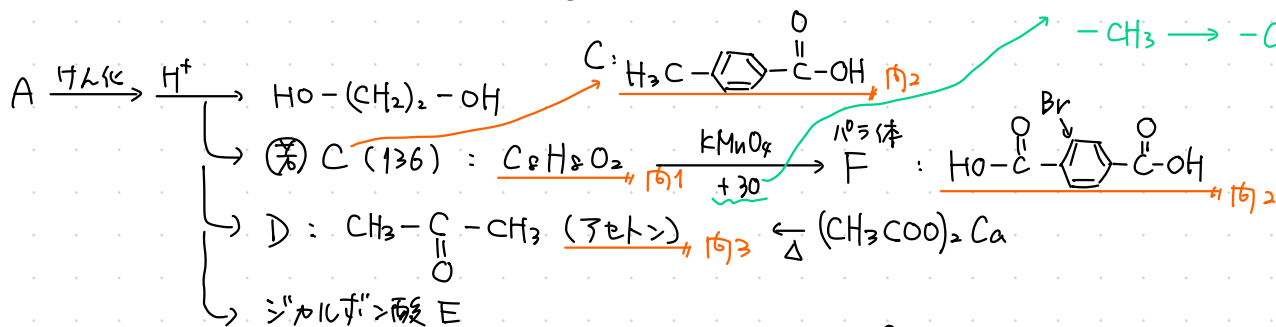
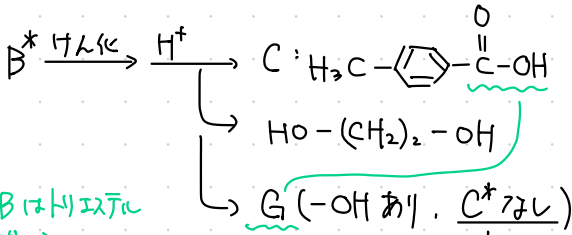
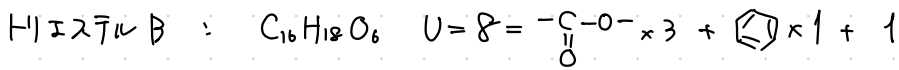


1 THU 2025

エステル A : $C_{16}H_{18}O_6$ $U=8 = -\overset{\text{O}}{\parallel}{C}-O- \times 3 + \text{C}_6\text{H}_5 \times 1 + 1$

$\xrightarrow{KMnO_4}$ $2^\circ + 30$ は
 $-CH_3 \rightarrow -COOH$

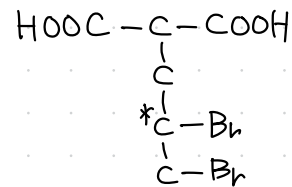
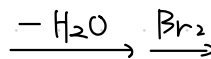
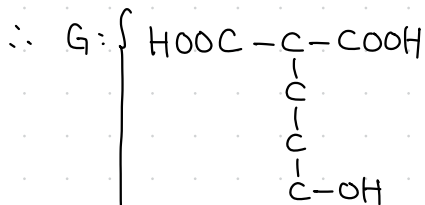
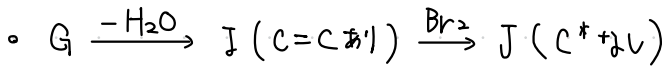
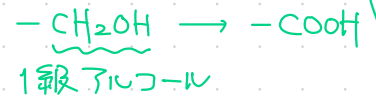
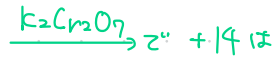
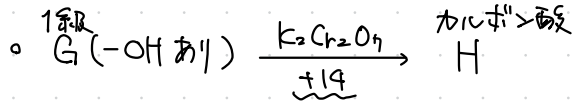




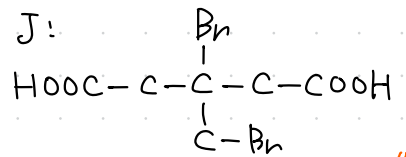
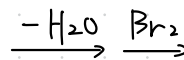
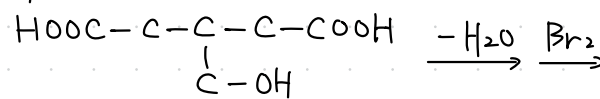
Bはトリエステル
だから,
Gはジカルボキ酸

• 対称面あり

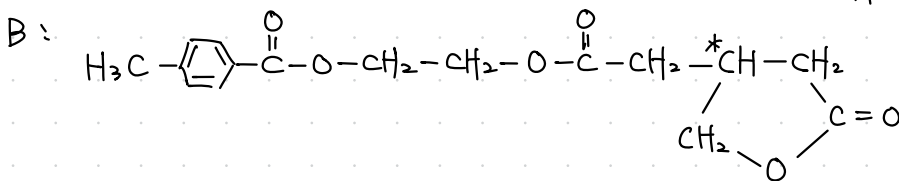
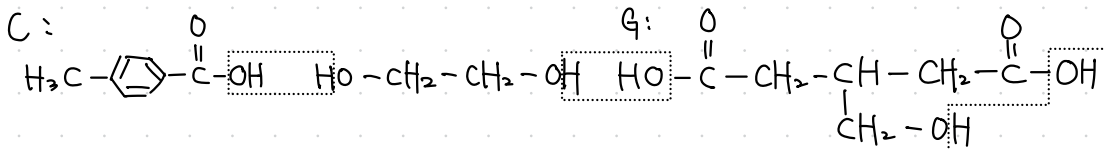
• $G = B^* + 3\text{H}_2\text{O} - (C + C_2\text{H}_6\text{O}_2) = C_6\text{H}_{10}\text{O}_5 \quad (U=2)$



G:



1676

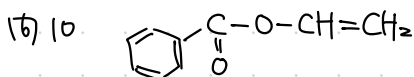


1677

環×1

1678 b. d

1679 田舎



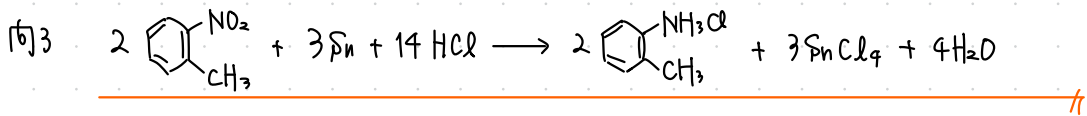
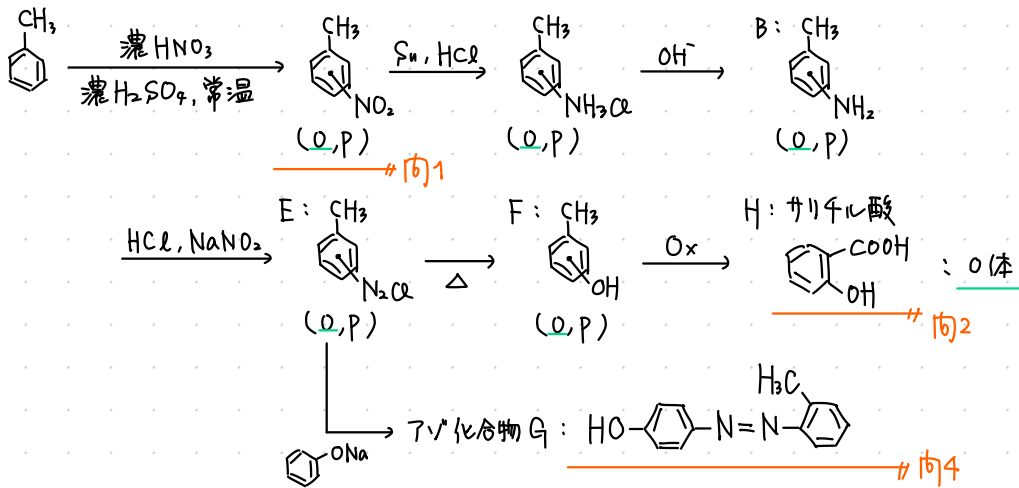
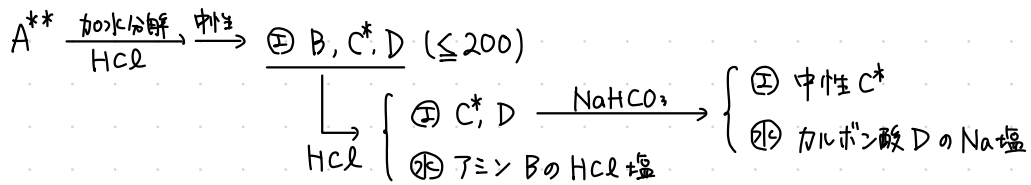
16711 2.5 L $\left(\frac{222 \text{ g}}{148 \text{ g/mol}} \times n = 0.60 \text{ mol/L} \times V_L \right)$

16712 (1) N: $\text{C}=\text{O} > \left(\leftarrow \text{H}^+ \right)$ M: PVA

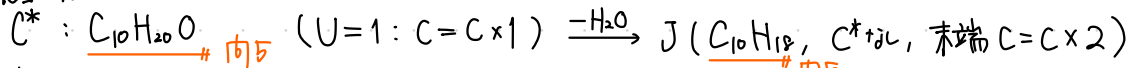
(2) $\frac{22 \text{ g}}{44 \text{ g/mol}} \times (44 + 6 \times 0.32)n \approx 23 \text{ g}$

2 THU 2024

$A^{**} : C_x H_y O_z N_w$, C^* 2017 4級C原子2117個.



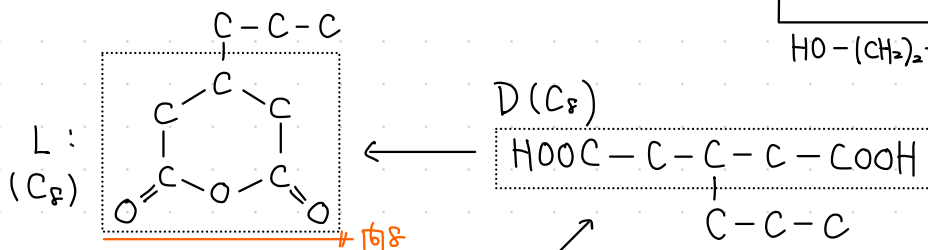
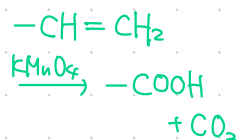
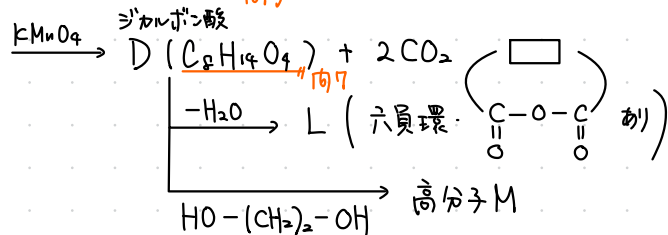
1級
アルコール



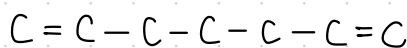
$\downarrow H_2$

K (C^*_{12} = 対称面あり)

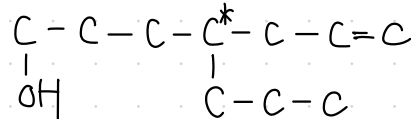
KMnO₄ ジカルボン酸



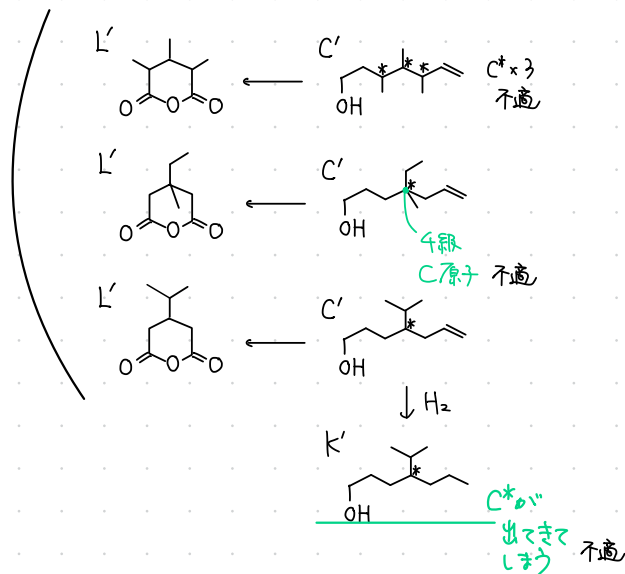
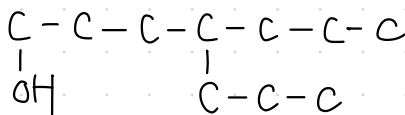
J (C_{10})



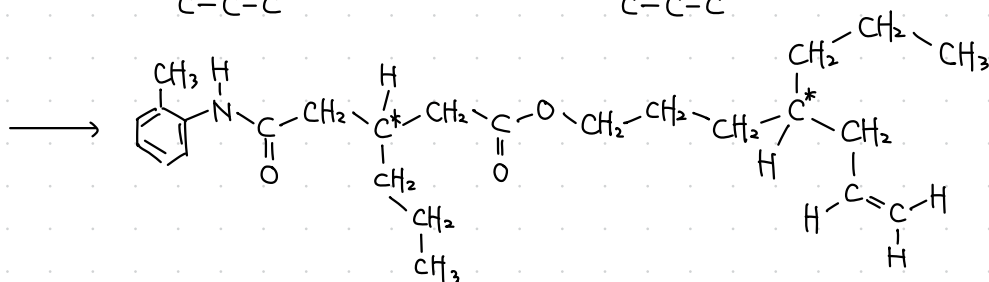
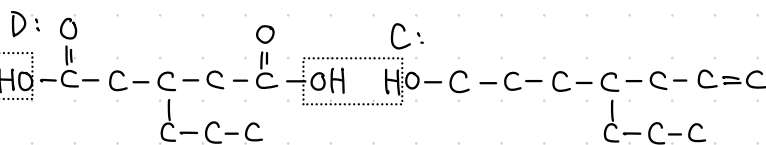
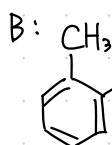
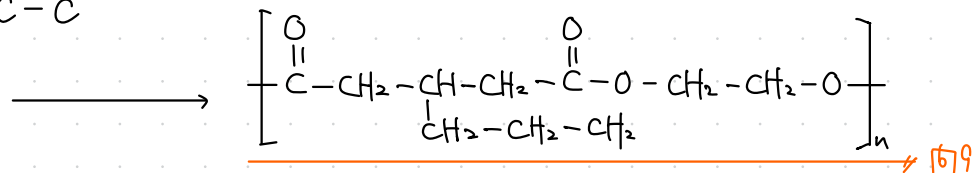
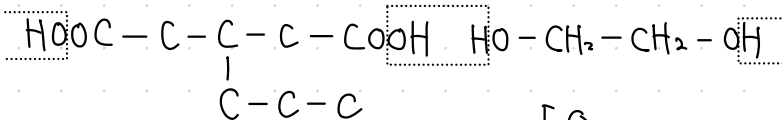
C (C_{10})



K (C^*_{12}) $\downarrow H_2$



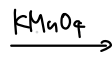
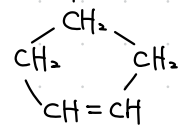
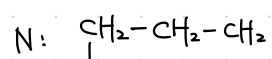
D:



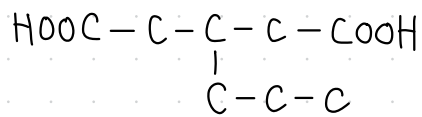
$\rightarrow$ 1612

16) 10 2.4×10^2 g

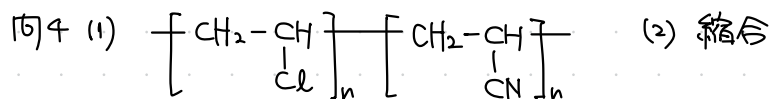
16) 11 C_8H_{14} $U=2$



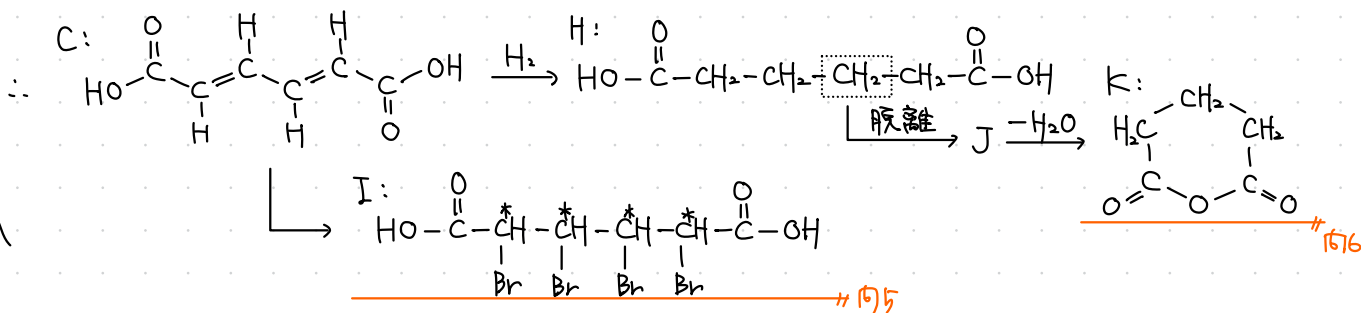
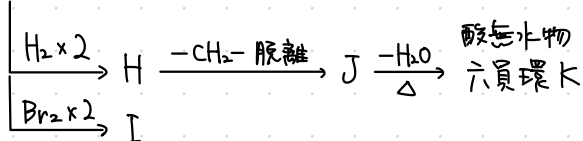
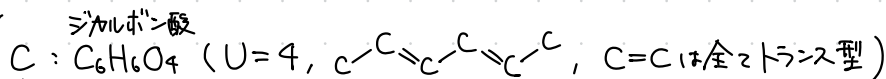
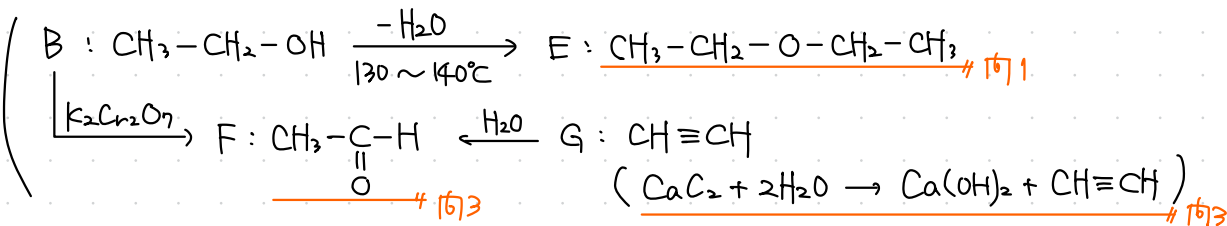
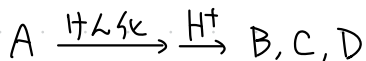
D:



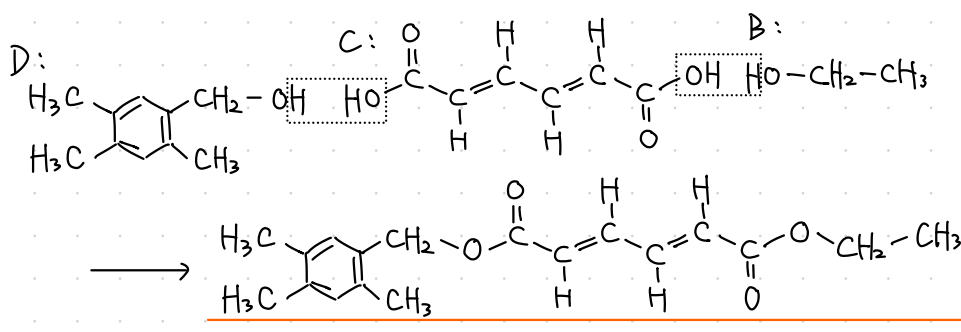
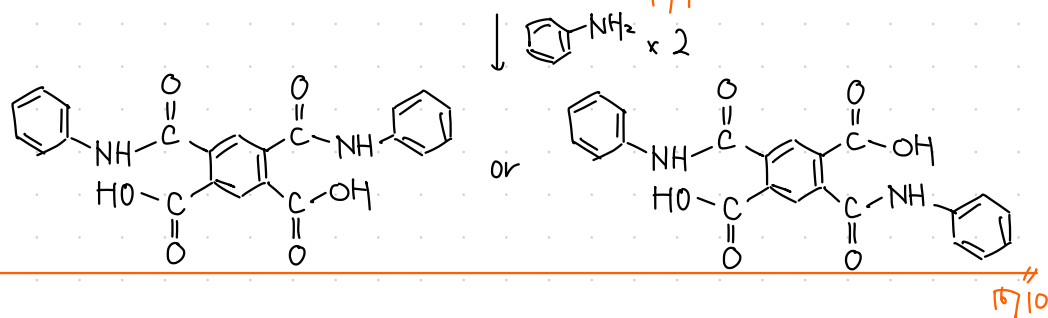
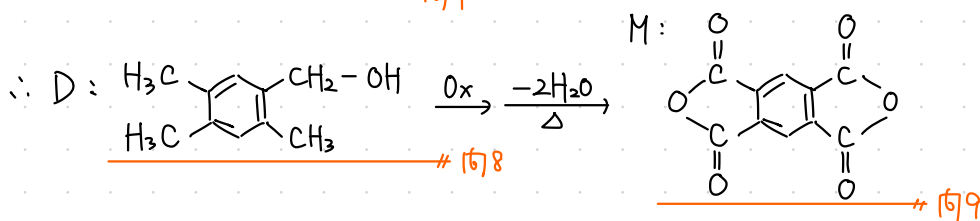
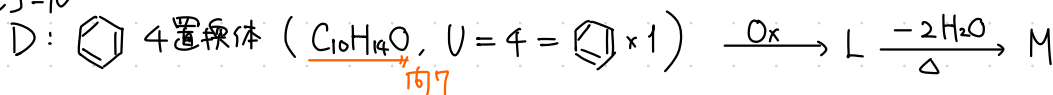
3 THU 2023



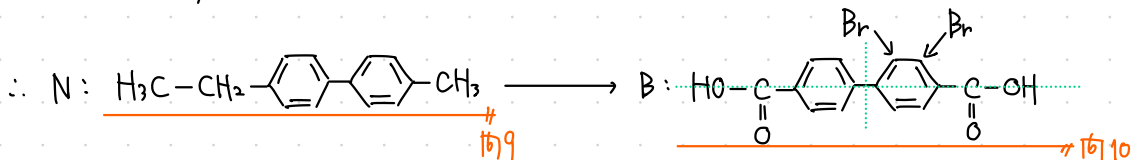
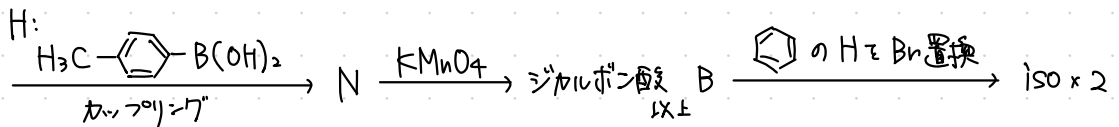
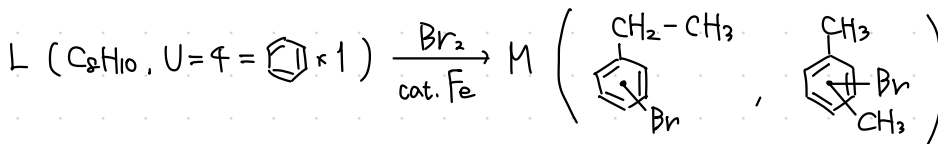
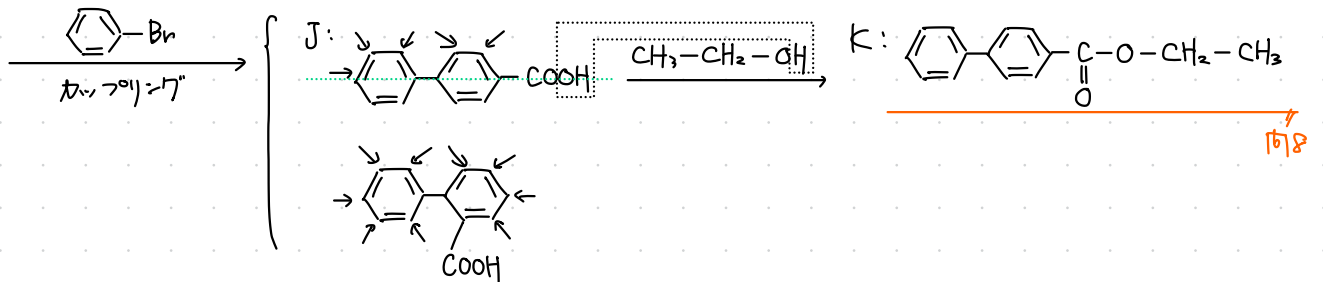
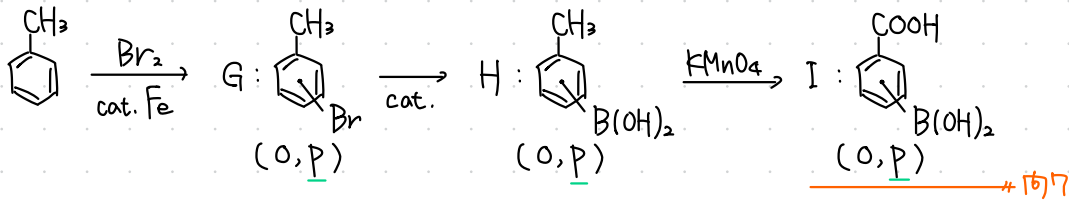
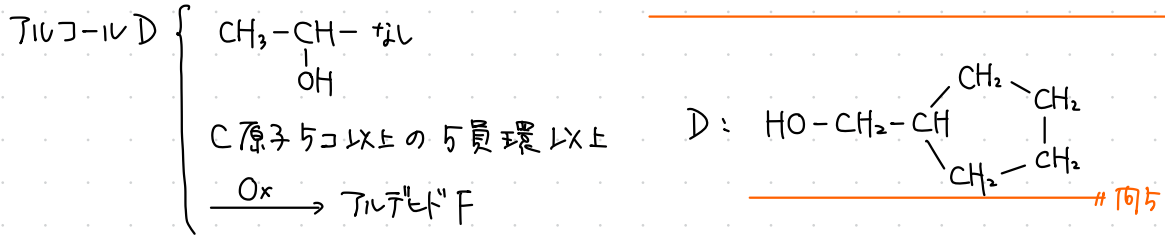
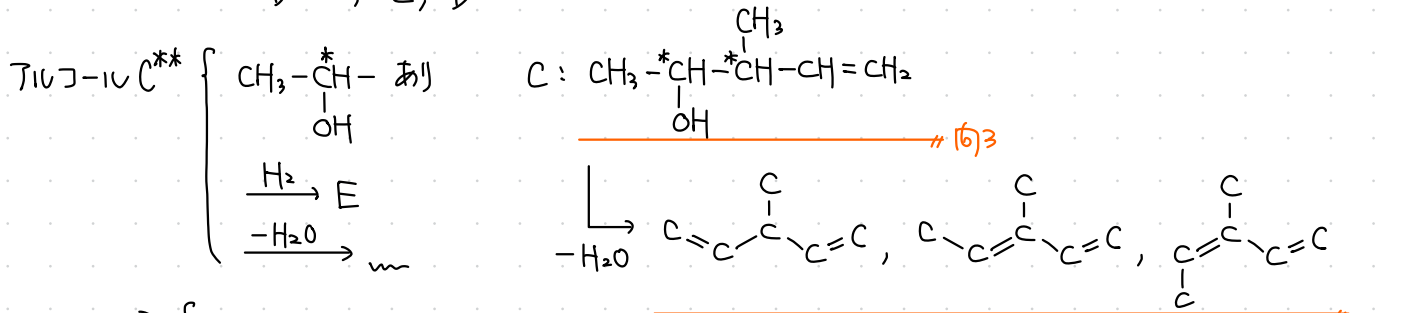
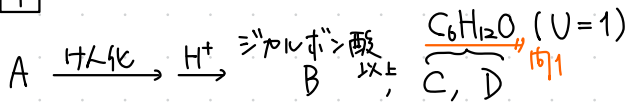
(3) $2.00 \times 10^2 : 1.00 \times 10^2$



第1級
アルコ-ル

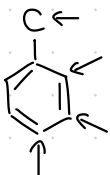


4



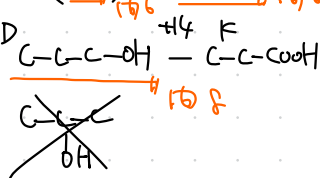
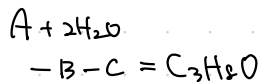
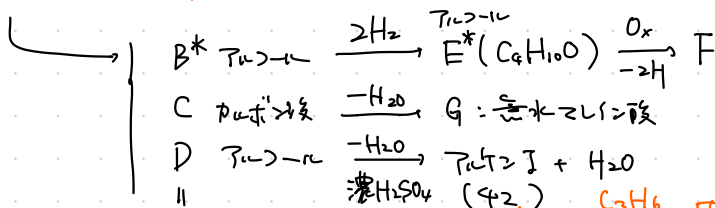
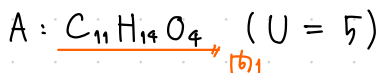
162 田舎

166

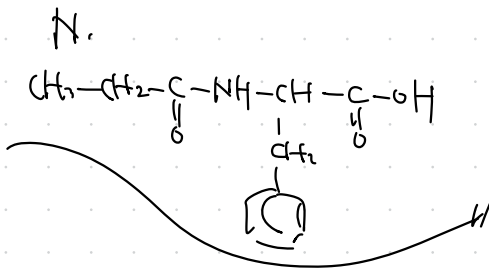


4

5

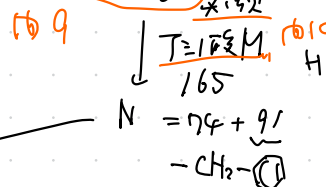
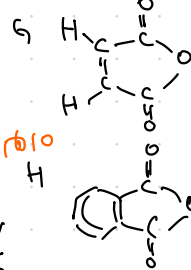
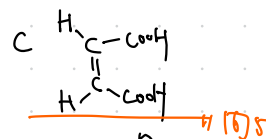
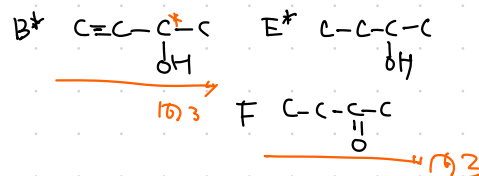
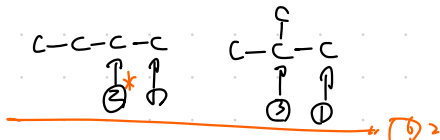


PP

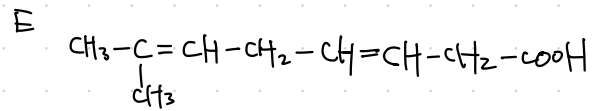
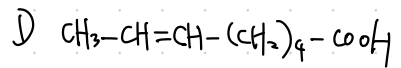
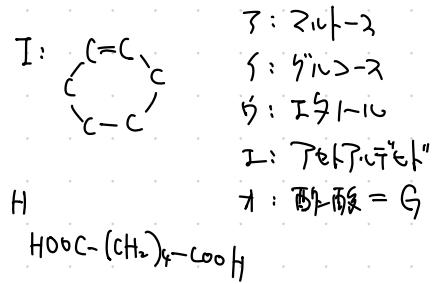
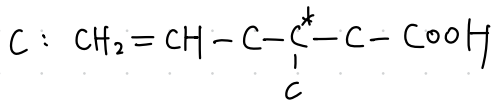
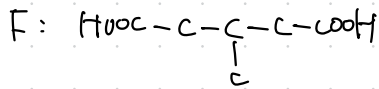
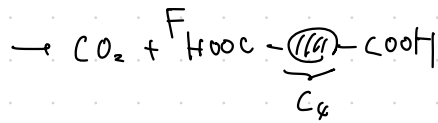
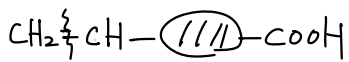
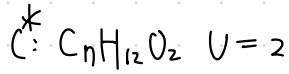
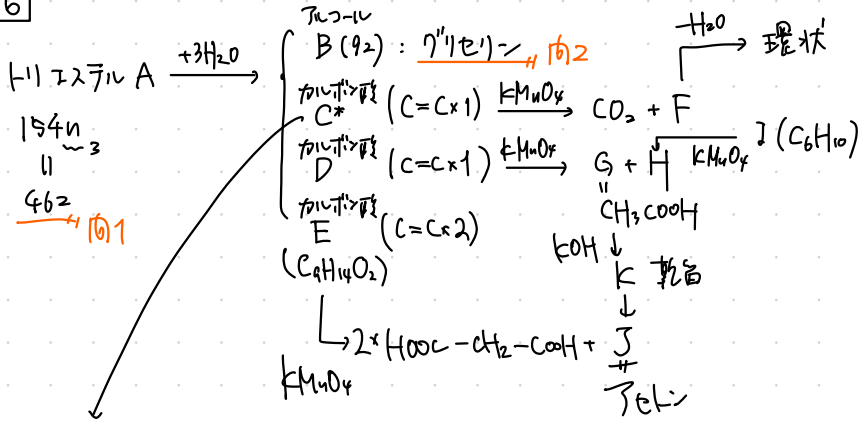


M: Phe

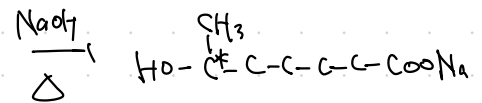
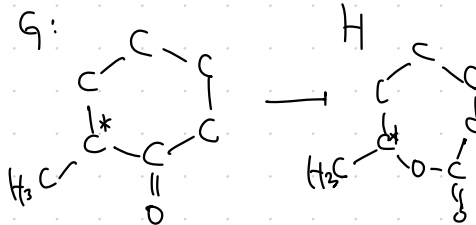
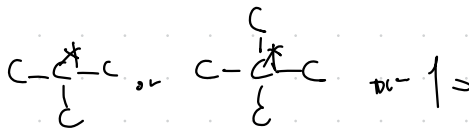
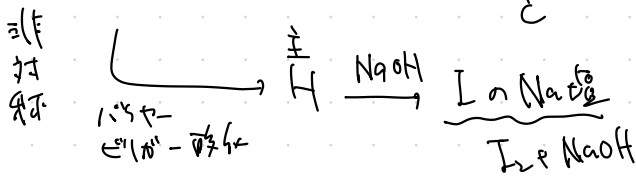
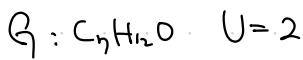
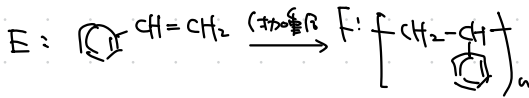
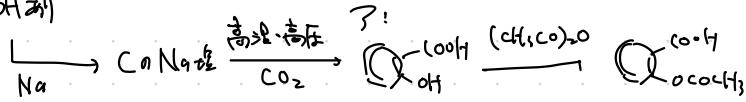
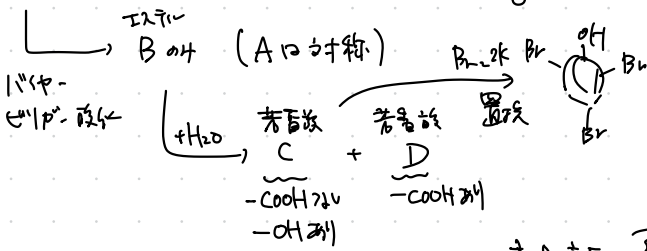
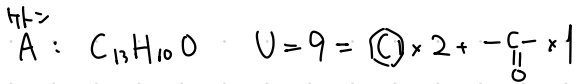
$N = 74 + 91$
 $-CH_2-$
 165



6

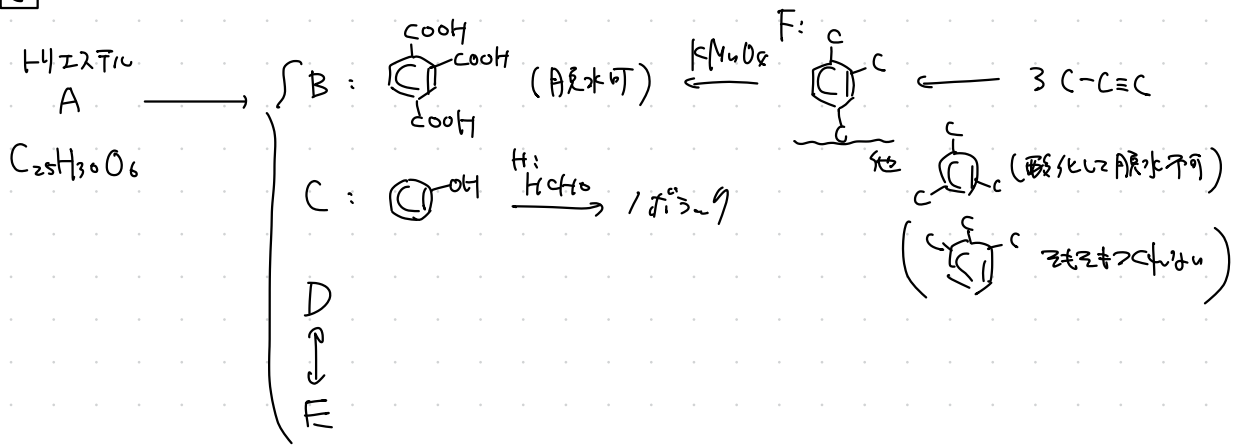


7

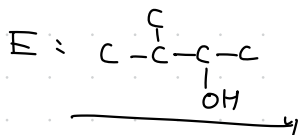
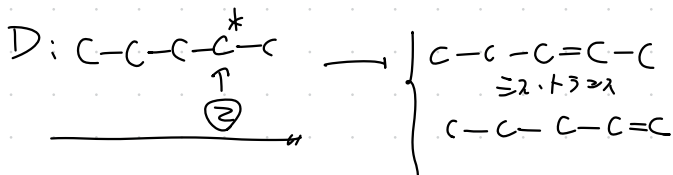


ア: e i: c り: h i: b

8

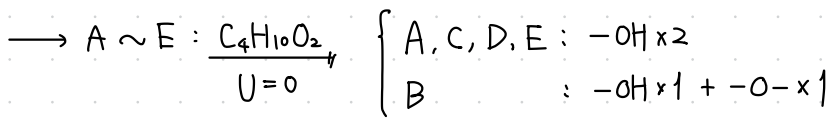


$$\text{D, E: } \frac{\text{C}_{25}\text{H}_{30}\text{O}_6 + 3\text{H}_2\text{O} - \text{C}_9\text{H}_6\text{O}_6 - \text{C}_6\text{H}_6\text{O}}{2} = \text{C}_5\text{H}_{12}\text{O}$$

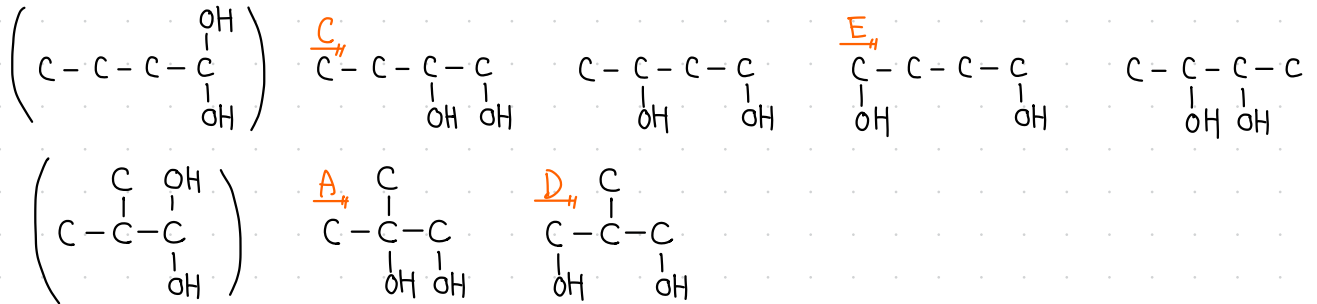


9

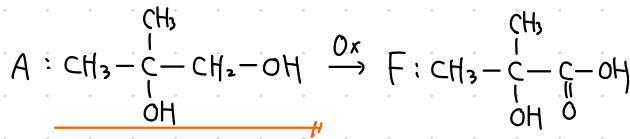
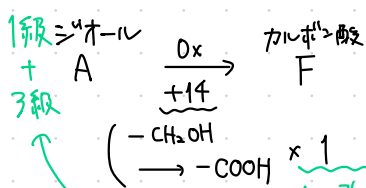
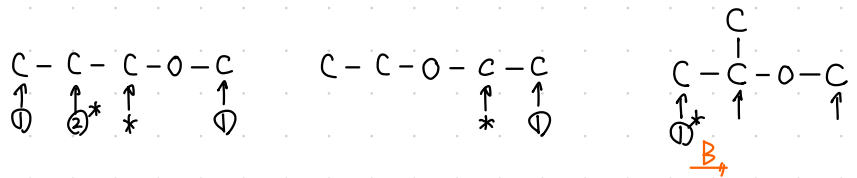
$$(C_2H_5O)_n \rightarrow \text{H数} \leq \frac{2 \cdot 2n + 2}{\text{最大H数}} \therefore n \leq 2 \quad \text{H数は偶数} \therefore n=2$$



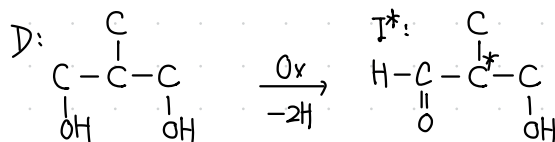
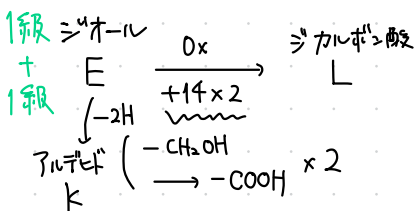
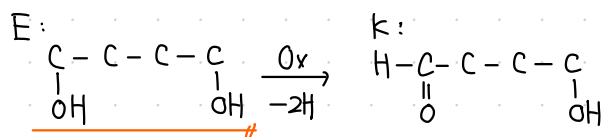
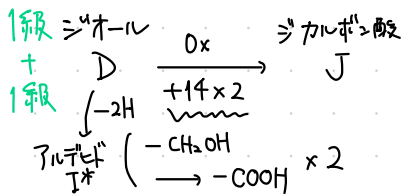
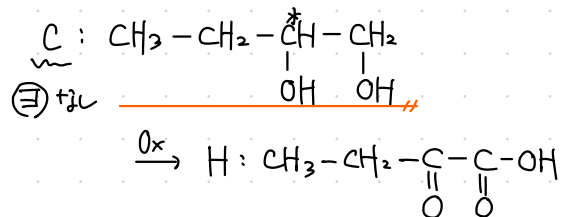
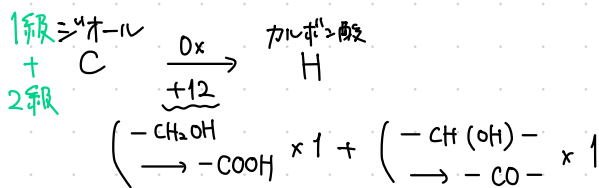
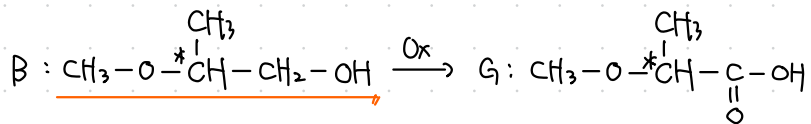
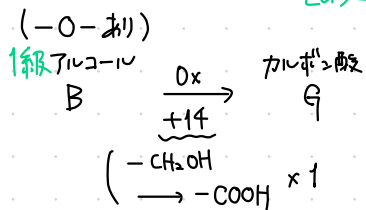
< A, C, D, E : -OH x 2 >

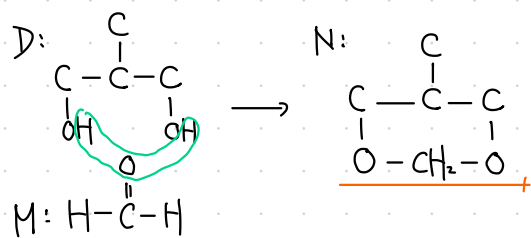


< B : -OH x 1 + -O- x 1 >

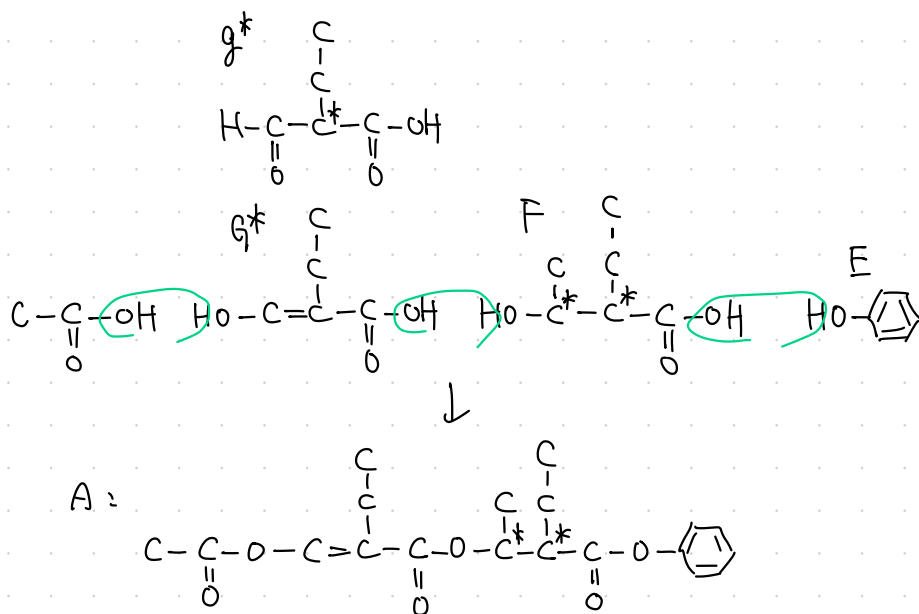
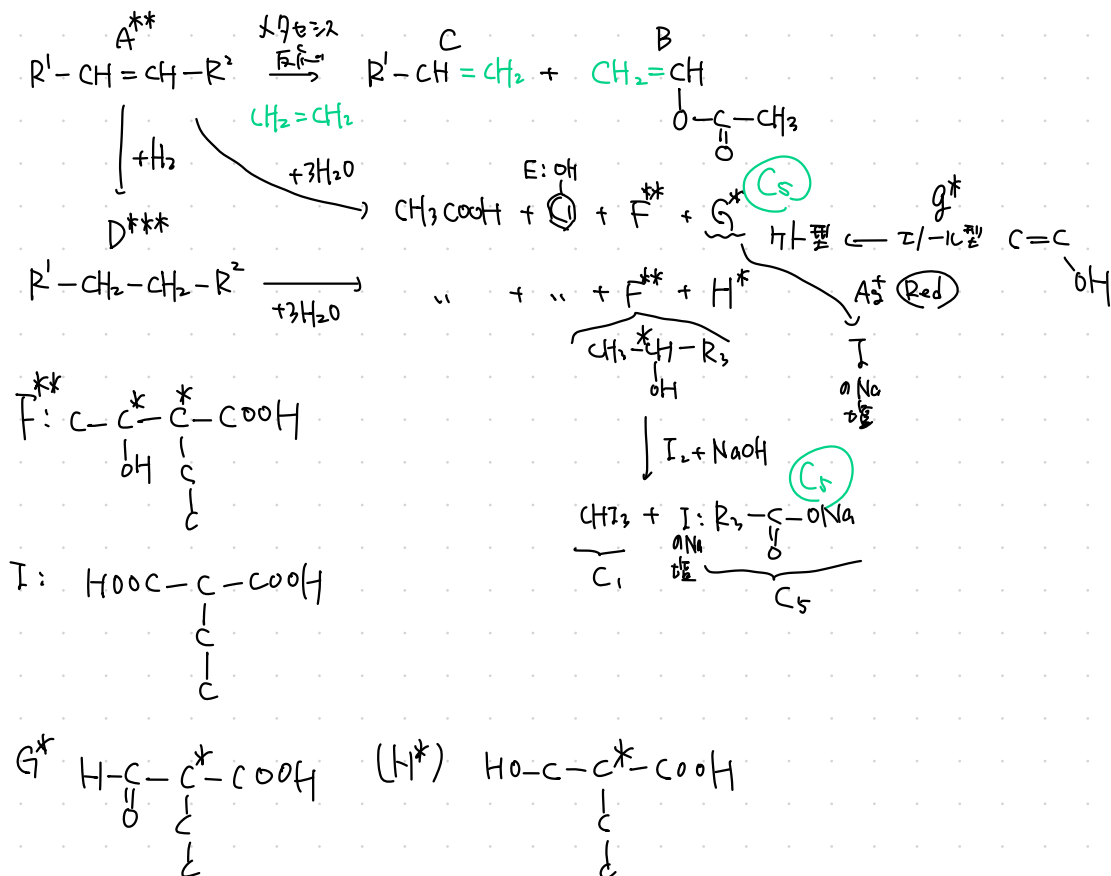
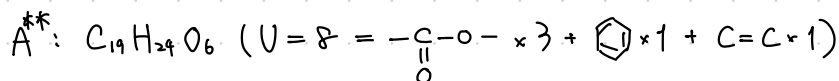


1か所酸化で済む
とゆきには、3級アルコール構造あり

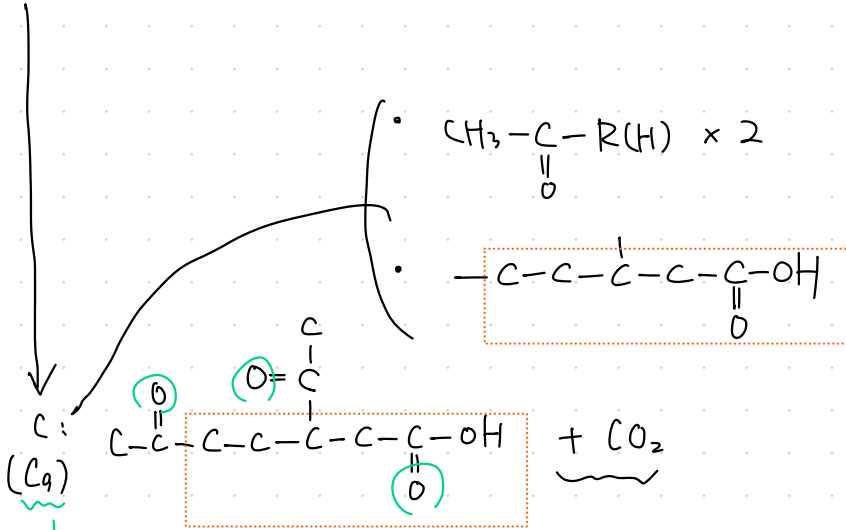
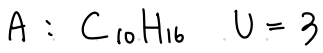




10 449



16



○ のうち 2ヶ所は 分子内 $\begin{cases} \text{C}=\text{C} \\ \text{環} \end{cases}$ $(U=2)$ + の2ヶ所 末端 $-\text{C}=\text{CH}_2$ $(U=1)$

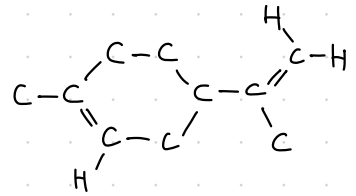
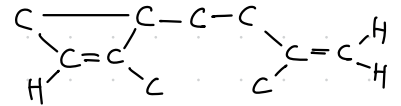
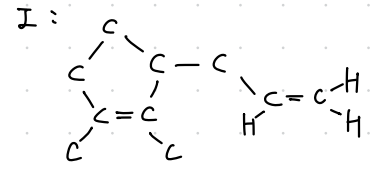


1: 二重結合 2つ

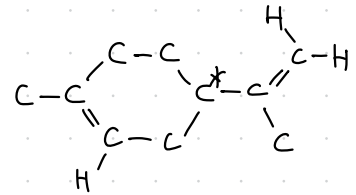
環 1つ

≡ 重結合 1つ

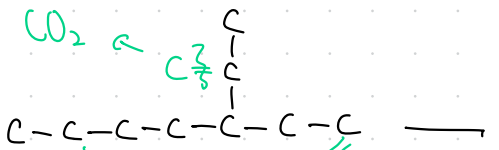
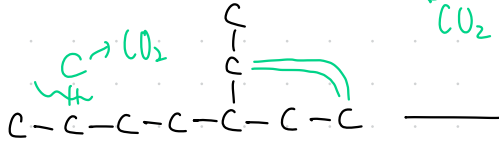
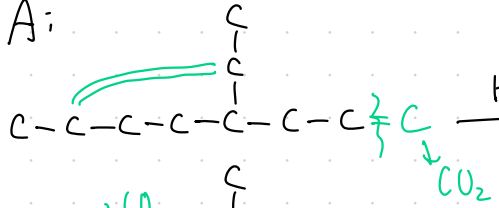
ウ: 3 4 3 3



オ



A:



B

